ADVANCED COMPUTER SCIENCE ASSESSMENT

Comprehensive Exam Paper | MCQs, MSQs & True/False with Explanations

Time Allowed: 45 Mins
Total Marks: 50

Candidate Name: _____ **Roll Number:** _____

Section A: Multiple Choice Questions (MCQs)

Question 1: Which of the following data structures operates on a Last-In-First-Out (LIFO) principle?

A) Queue

B) Stack

C) Array

D) Binary Tree

Answer: B) Stack

Explanation: A stack is a linear data structure that follows the Last-In-First-Out (LIFO) model where elements are added and removed exclusively from the top.

Question 2: What is the worst-case time complexity of QuickSort algorithm?

A) O(N log N)

B) O(N)

C) $O(N^2)$

D) O(log N)

Answer: C) $O(N^2)$

Explanation: QuickSort exhibits $O(N^2)$ time complexity in the worst-case scenario when the chosen pivot consistently results in highly unbalanced partitions.

Section B: Multiple Select Questions (MSQs)

Question 3: Which of the following protocols operate at the Transport Layer of the OSI model? (Select all that apply)

[ ] A) TCP

[ ] B) IP

[ ] C) UDP

[ ] D) HTTP

Answer: A, C (TCP and UDP)

> **Explanation:** Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) are core transport layer protocols, whereas IP operates at the Network layer and HTTP at the Application layer.

Section C: True or False Questions

Question 4: HTTPS encrypts HTTP traffic using SSL/TLS to ensure secure communication over a network.

(True / False)

Answer: True

> **Explanation:** HTTPS wraps standard HTTP requests and responses within an encrypted TLS layer to safeguard user data from eavesdropping.

Question 5: In object-oriented programming, multiple inheritance is natively supported without ambiguity in Java.

(True / False)

Answer: False

> **Explanation:** Java does not support multiple inheritance with classes to avoid the Diamond Problem, though it achieves similar behavior using interfaces.